MOHAMMAD FAROOQUE AHMED

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## EDUCATION

**Kalinga Institute of Industrial Technology** 2023 – Present
*B.Tech in Information Technology* *Bhubaneswar, Odisha*

## EXPERIENCE

**Intern at Tata Steel** May – June 2025
*Automation Division* *Jamshedpur, Jharkhand*

- Performed exploratory data analysis on HR and Google Play datasets (10k+ records), identifying key behavioral and business trends.
- Developed a Linear Regression model on California Housing dataset (20k+ rows), achieving $R^2$ 0.60.
- Automated Excel-to-Python data workflows, reducing manual analysis time by 30%.
- Generated 5+ actionable insights through data visualization and statistical techniques to support business decisions.

## TECHNICAL SKILLS

**Languages:** Python, C/C++, Java, SQL

**AI/ML:** NLP, RAG, Neural Networks, LLMs

**Frameworks:** LangChain, HuggingFace, OpenCV

**Tools:** Power BI, MySQL, Jupyter Notebook, Excel, GitHub

## PROJECTS

**Student Performance Prediction** | Live Flask, CatBoost, AWS

- Developing an end-to-end ML web application using Flask to predict student performance from structured datasets (1,000+ records).
- Optimizing CatBoost regression model with hyperparameter tuning, reducing RMSE from 14.0 to 4.57 (67% improvement).
- Designing modular data pipelines with logging and exception handling to improve maintainability and debugging efficiency by 30%.
- Deploying the application on AWS for scalable, real-time prediction services.

**Network Security System, MLOps Pipeline** *In Progress*
MLflow, DagsHub, Docker, MongoDB, AWS

- Building a production-grade end-to-end MLOps pipeline (ETL → validation → training) handling 50k+ records using MongoDB.
- Implementing CI/CD pipelines with GitHub Actions and Docker, improving deployment efficiency by 40%.
- Tracking experiments and optimizing models using MLflow, achieving 25% reduction in prediction error.
- Deploying scalable REST APIs on AWS EC2 and S3 with sub-200ms latency for real-time inference.

**Khetify — AI Crop Recommendation System** | Live Python, Flask, Random Forest, REST API, Render

- Engineered and deployed an end-to-end ML web application predicting the optimal crop from 22 classes using 7 soil and climate features, serving real-time results via a REST API.
- Trained a Random Forest Classifier (200 estimators) on 2,200+ agronomic samples, achieving ∼99% test accuracy and ∼99% cross-validated accuracy (±0.002), validated on a held-out 440-sample test set.
- Built a complete data pipeline — StandardScaler normalization, Label Encoding, and Pickle-based model serialization — enabling sub-millisecond inference with Top-3 ranked crop predictions and confidence scores.
- Conducted feature importance analysis identifying rainfall (28%) and humidity (22%) as dominant predictors; deployed a mobile-responsive Flask app on Render, accessible globally.